Experiment 7: Random Sampling and Sampling Distribution

Aim:

To explore random sampling from a population and understand the concept of sampling distribution using Python in Jupyter Notebook.

Steps:

1. Generate a Population:

○ Create a population of data with a specified distribution (e.g., normal distribution).

2. Random Sampling:

○ Perform random sampling from the population to create multiple samples of different sizes.

○ Compute sample statistics (mean, standard deviation, etc.) for each sample.

3. Sampling Distribution:

○ Plot histograms or density plots of sample statistics (e.g., sample means).

○ Compare the sampling distribution of the sample statistic (mean) with the population distribution.

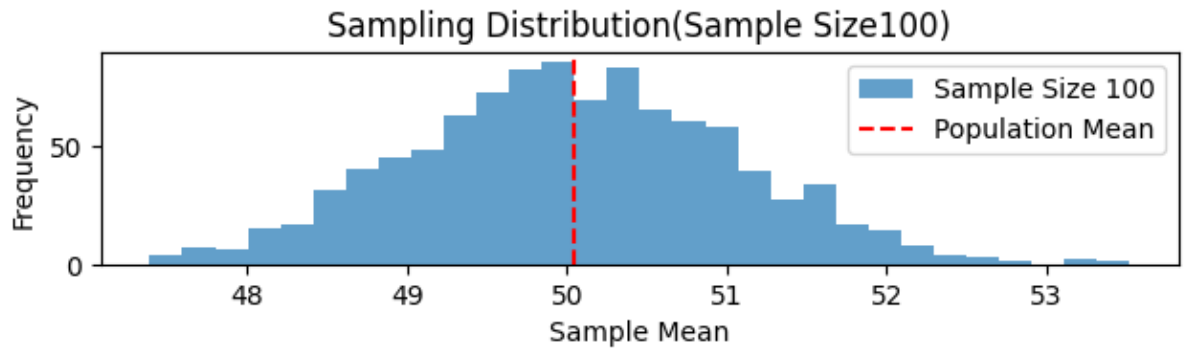
4. Central Limit Theorem (Optional):

○ Demonstrate the Central Limit Theorem by showing that as sample size increases, the sampling distribution of the sample mean approaches a normal distribution regardless of the population distribution.

Example Code:

```
import numpy as np
import matplotlib.pyplot as plt
population_mean = 50
population_std = 10
population_size = 100000
population = np.random.normal(population_mean,population_std,population_size)
sample_size = [30,50,100]
num_samples = 1000
sample_means = {}
for size in sample_size:
 sample_means[size]=[]
 for _ in range(num_samples):
  sample = np.random.choice(population,size=size,replace=False)
```

```
    sample_means[size].append(np.mean(sample))
plt.figure(figsize=(12,8))
for i,size in enumerate(sample_size):
 plt.subplot(len(sample_size),1,i+1)
 plt.hist(sample_means[size],bins=30,alpha=0.7,label=f'Sample Size {size}')
 plt.axvline(np.mean(population),color='red',linestyle='dashed',linewidth=1.5,label="Population Mean")
 plt.title(f'Sampling Distribution(Sample Size{size})')
 plt.xlabel('Sample Mean')
 plt.ylabel('Frequency')
 plt.legend()
 plt.tight_layout()
 plt.show()
```



Result:

Thus the concept of random sampling and sampling distribution understand by using python in jupyter notebook.